



ODONT 541.1

CARIOLOGY

Diagnostic Techniques

Lecture #7

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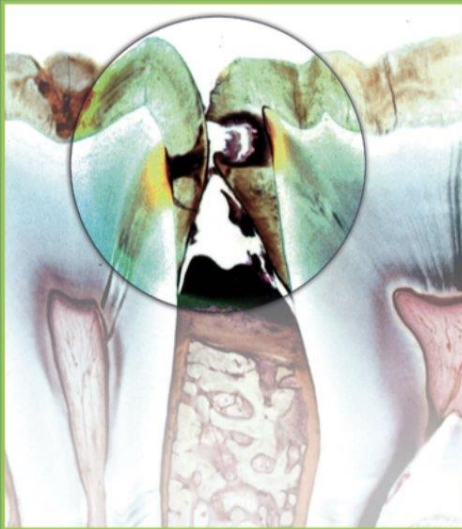
Office Hours: By appointment
(Tuesday and Wednesday afternoons)

Mission: To advance the science and art of health through education, service, scholarship and social accountability

Fourth Edition

Dental Caries

The Disease and Its Clinical Management



Edited by **Ole Fejerskov** and **Bente Nyvad**

WILEY Blackwell

Textbook Pages 61- 100

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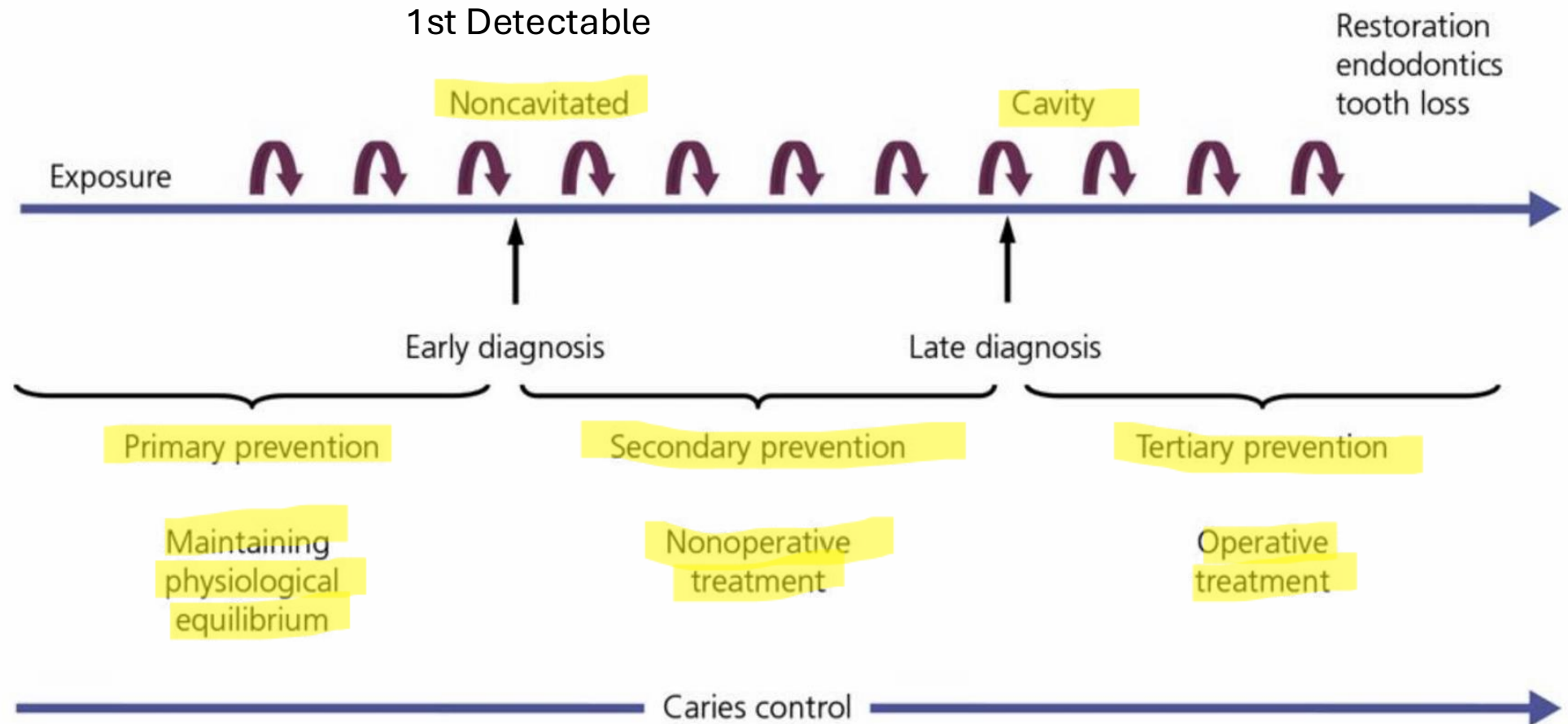


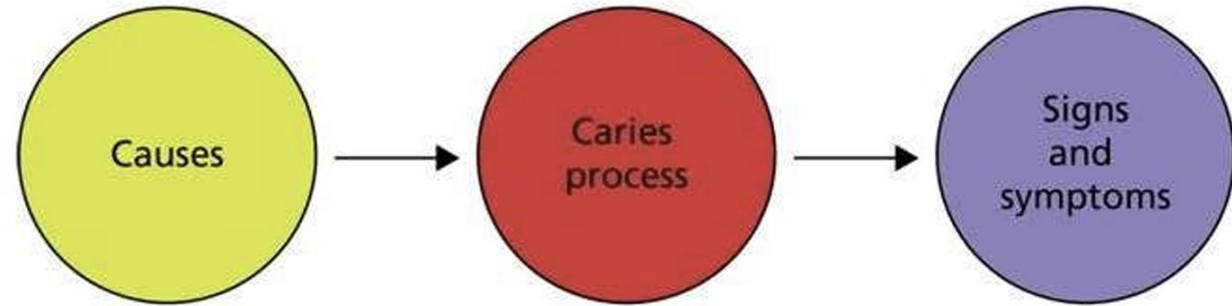
Figure 13.1 Schematic illustration of the caries control concept. Because of continuous exposure to the metabolically active biofilm, disease control must be maintained lifelong. Both nonoperative and operative treatments are part of the caries control concept, but operative treatments should never be the only treatment provided for patients with active caries lesions. See text for a detailed explanation.

When does caries start?

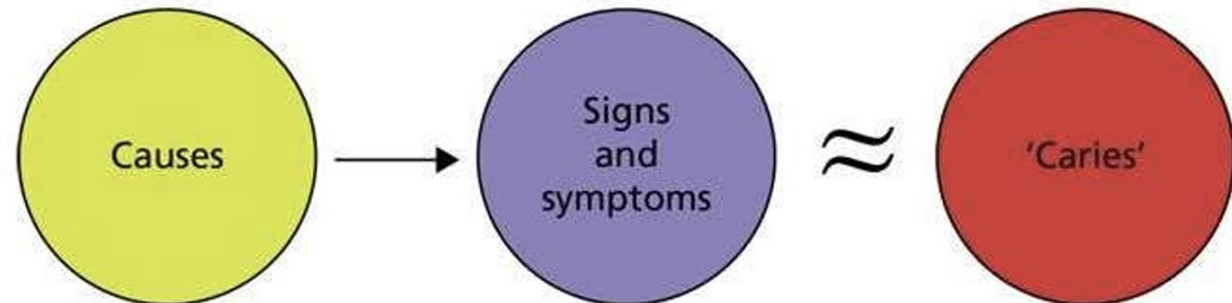
(a) Essentialistic concept: The causes (a caries "truth") result in a disease that manifest as signs and symptoms.

(b) Nominalistic concept: The disease name is no more than a label given to certain characteristics of the signs

Essentialistic caries concept



Nominalistic caries concept



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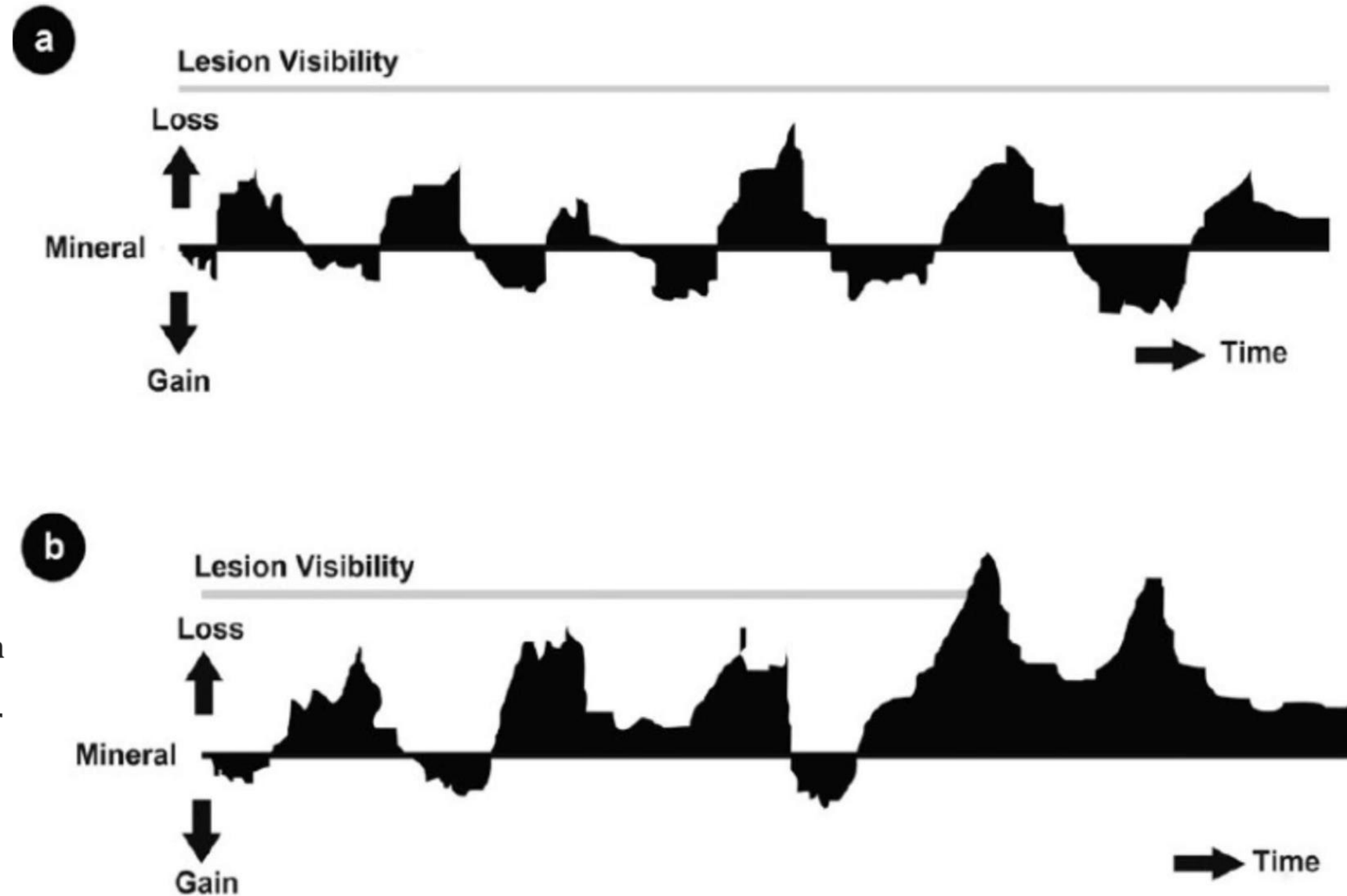
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- The essentialistic view pursues a search for the caries ‘truth,’
 - Historically dominated caries diagnostic research and testing method assessments
 - Modern concept of caries renders this search for a "gold standard" **inadequate**
- The **nominalistic** view is more concerned with making diagnoses that reflect the best caries management options.
 - The point of caries diagnosis is to categorize caries lesions to reflect the **best management** options, which in turn necessitates a profound understanding of the caries causal processes and how they may be interfered with.

A= Background demin/remin that stays below visual caries threshold

B = imbalance in demin vs remin, resulting in a visible white spot lesion over time.

"The transformation of dental caries from a physiological process to pathology is **not a sudden cross-over, but a continuum** over a period of time, affected by numerous variables"

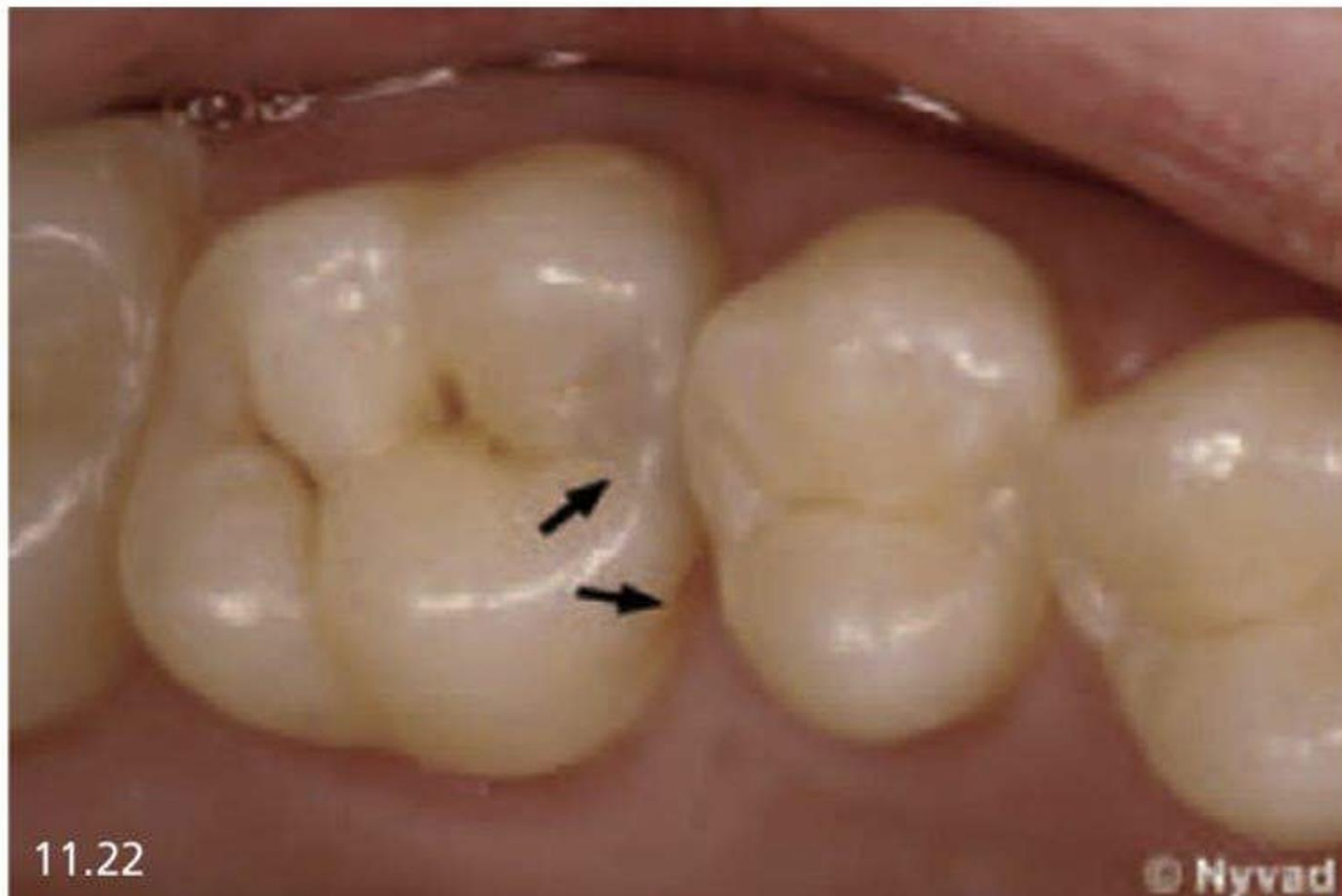


Detection Methods

- Pre 1900s- Visual tactile inspection only
- 1925- Bitewing radiography
- 1990s- Fluorescence

- Are periapical radiography and CBCT radiography diagnostic for caries?

Visual-Tactile



Visual-Tactile





Figure 15.7 Sequential stages of non-operative caries control in active cavitated lesions in primary teeth. The cavity in the first primary molar had previously been filled, but the filling was lost. (a) and (b) show active carious cavities before and after removal of unsupported enamel to make the cavities cleansable with a toothbrush. (c) and (d) show successive stages of lesion arrest in the second molar after three and six months, respectively. The floor of the cavity became harder and darker over time. Note that the first permanent premolar erupts into a clean environment!

Courtesy of Niels V. Hansen.



Figure 15.5 Consecutive changes of non-operative treatment of active cavitated root-surface caries lesions on the buccal surfaces of lower first and second premolars (Fig. 15.5a–d). The illustrations show the clinical appearance of the active lesions after 2, 4, and 10 years, respectively. The successful treatment was achieved through careful daily plaque removal with a fluoride toothpaste. After four years (c), an overhanging rim of unsupported enamel along the occlusal part of the lesion was removed to facilitate cleaning. Although cosmetically a problem to most patients, these lesions did not need operative treatment, which might weaken the teeth substantially and reduce their survival.

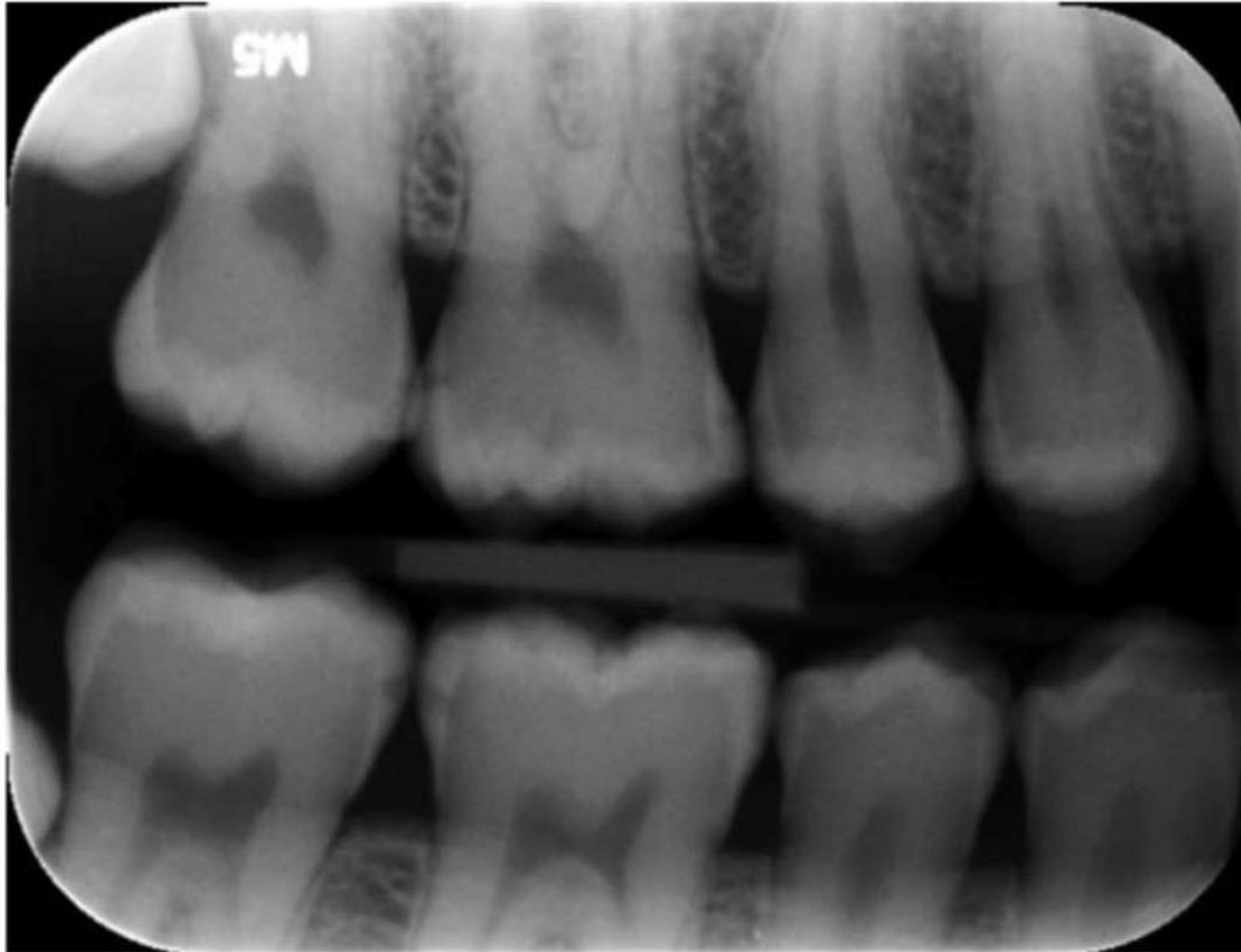
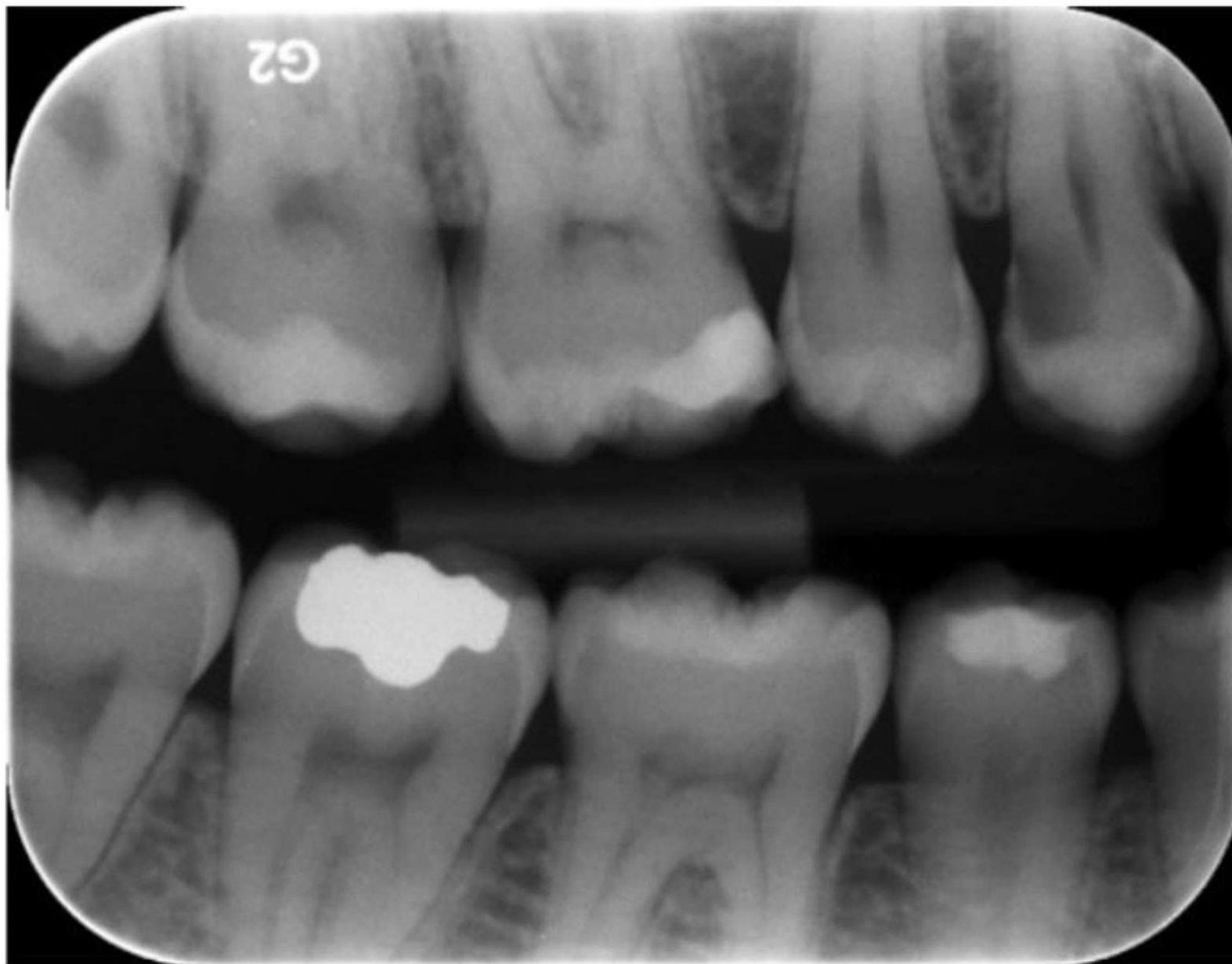


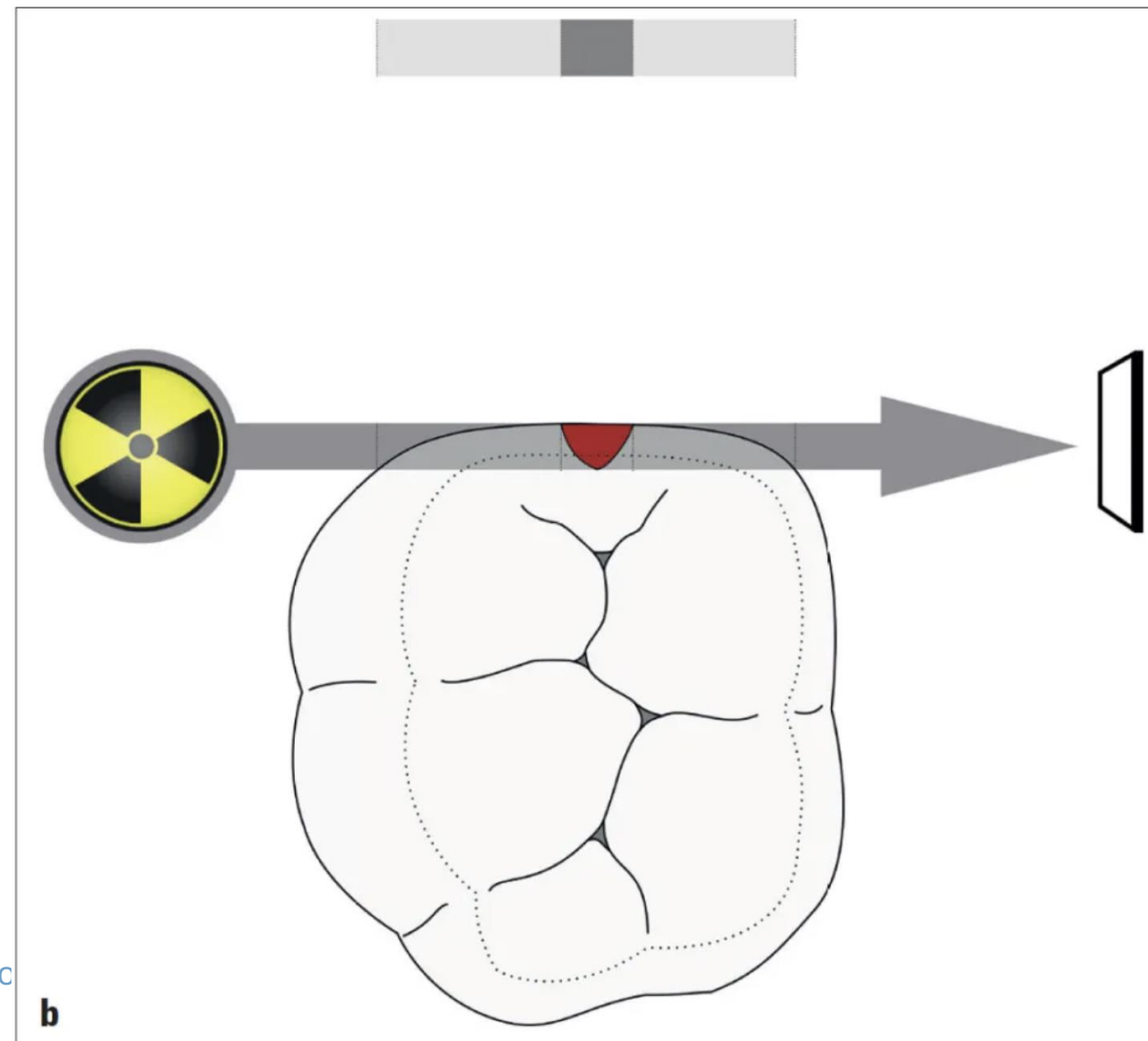
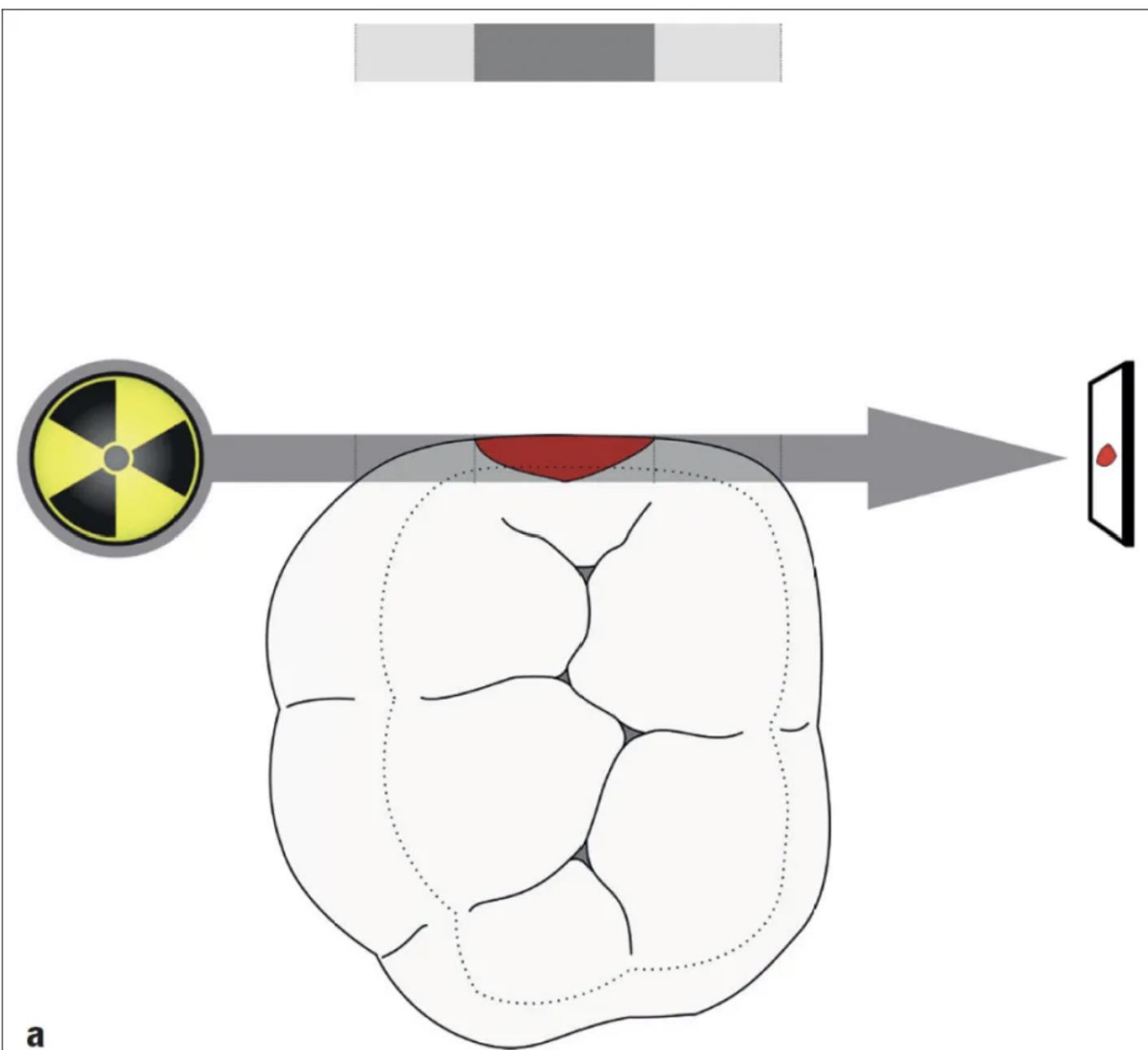
Figure 12.1 Bitewing radiograph showing radiolucencies in nearly all approximal surfaces due to caries. [social](#)



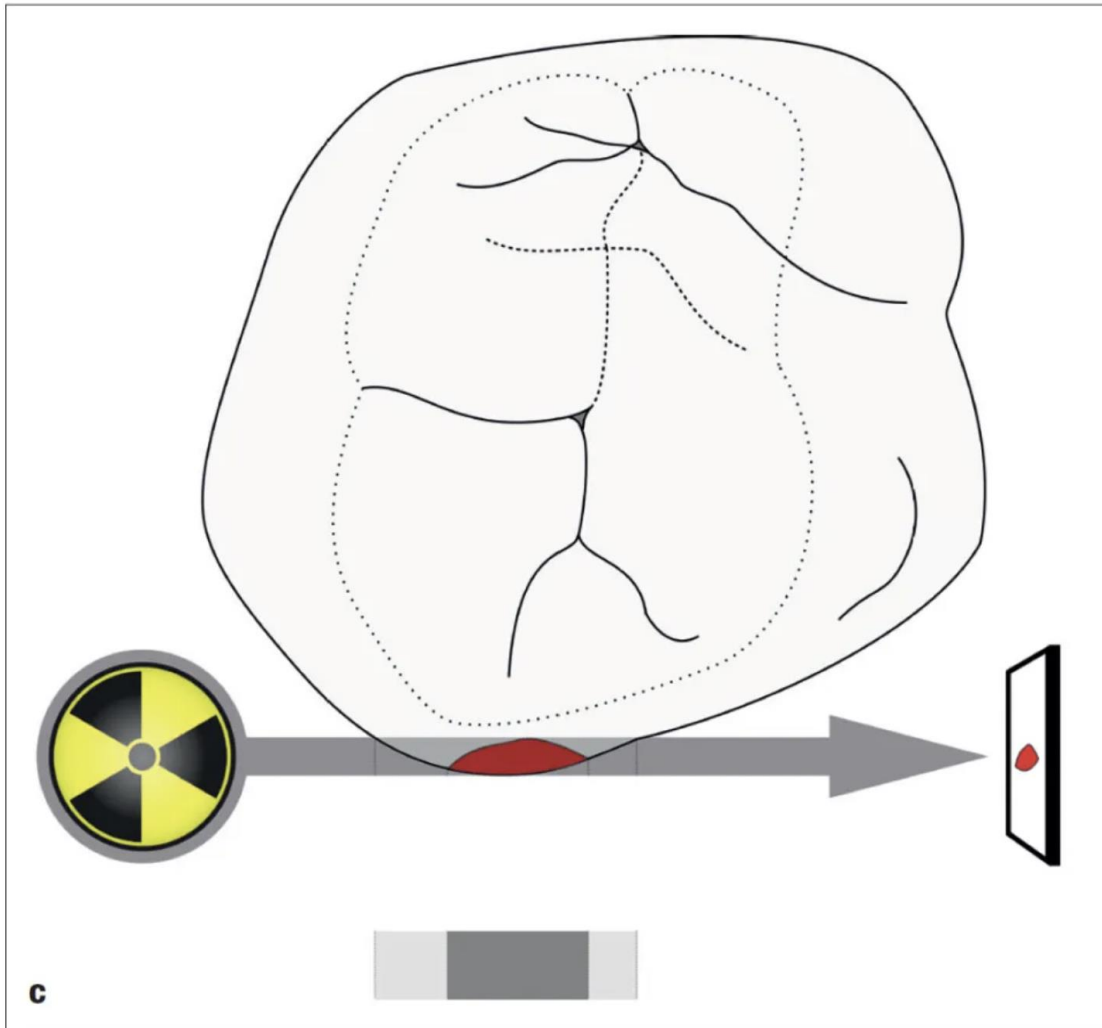
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FIG 3-31 (a to d) Diagram illustrating the rule of three thirds, ie, if the buccolingual lesion width is at least equal to one-third of total width, it will probably be visible on the radiograph. If the lesion width is less, it will be more difficult to see.

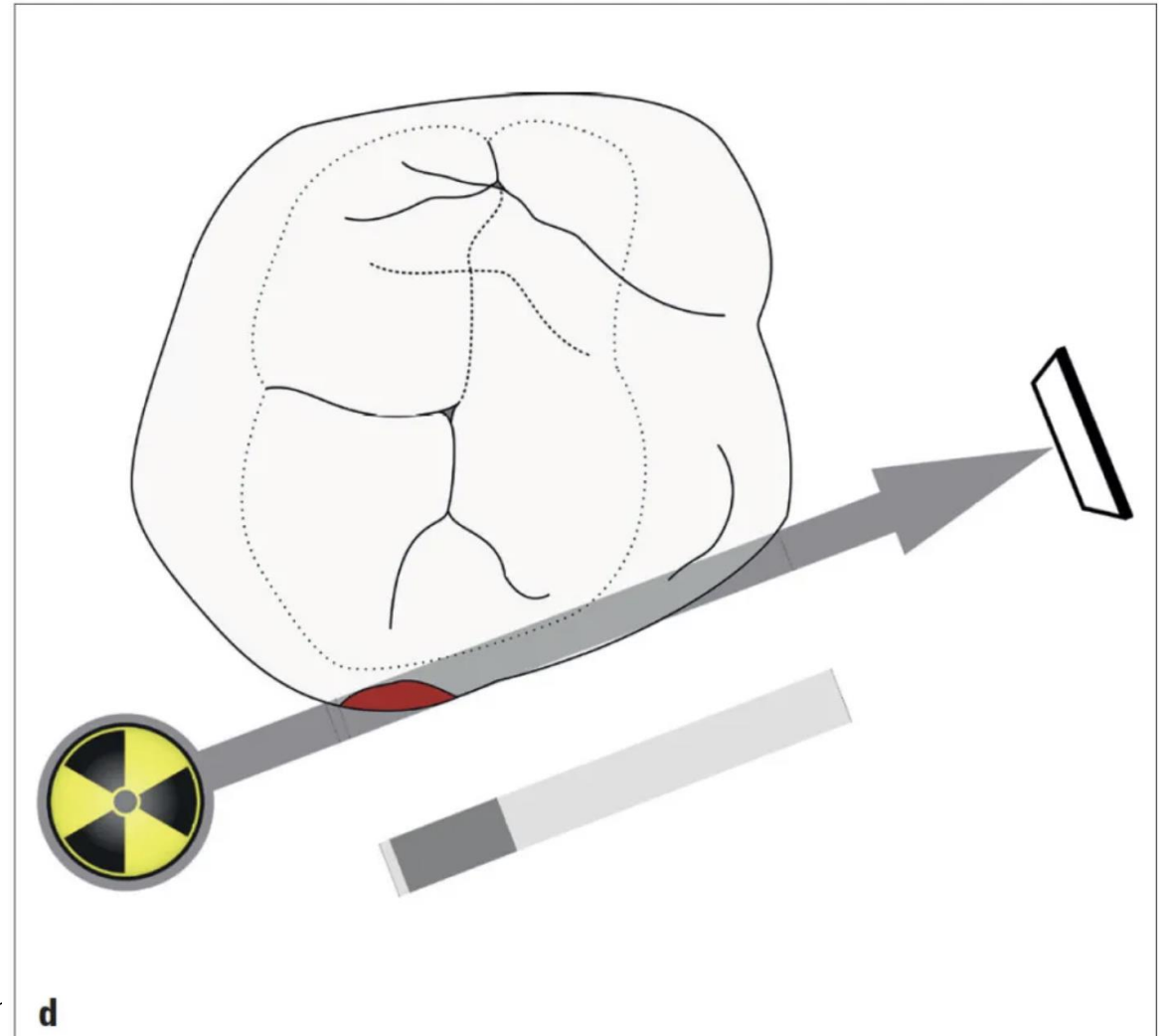


Beam angulation can also affect the radiographic detectability of a lesion.



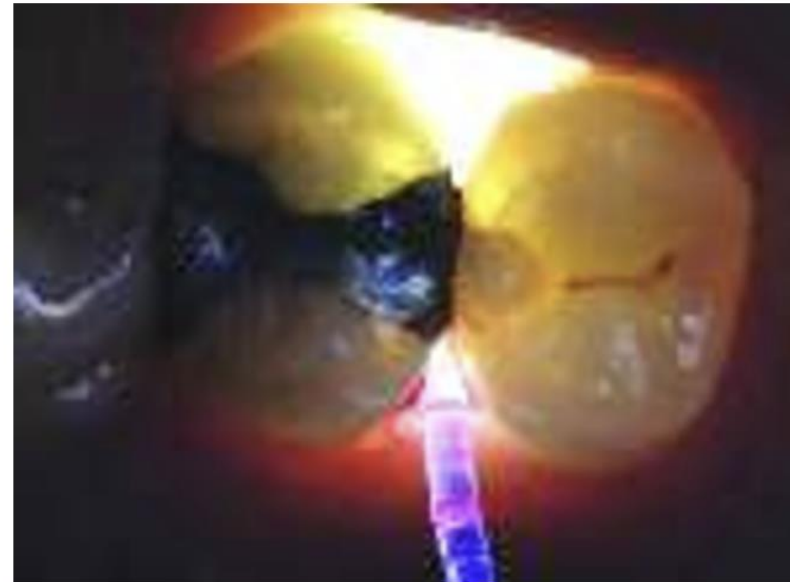
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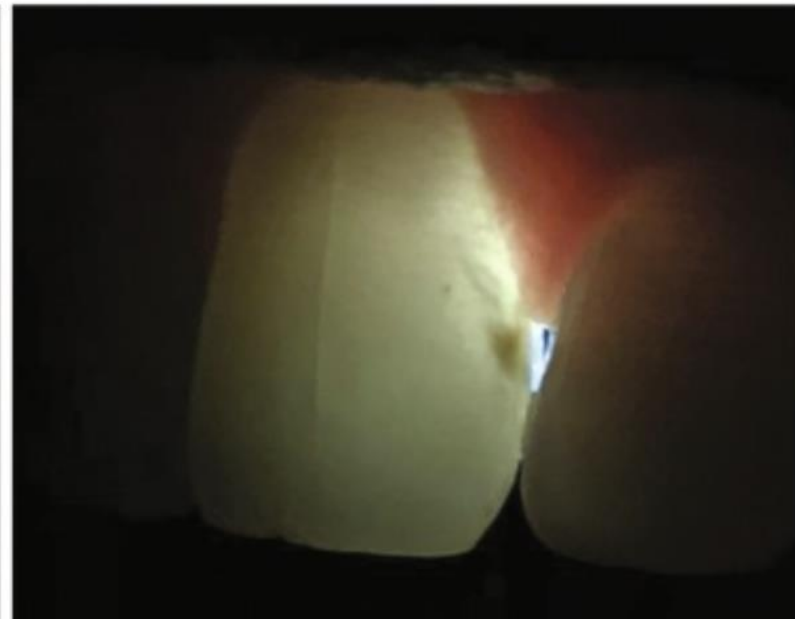
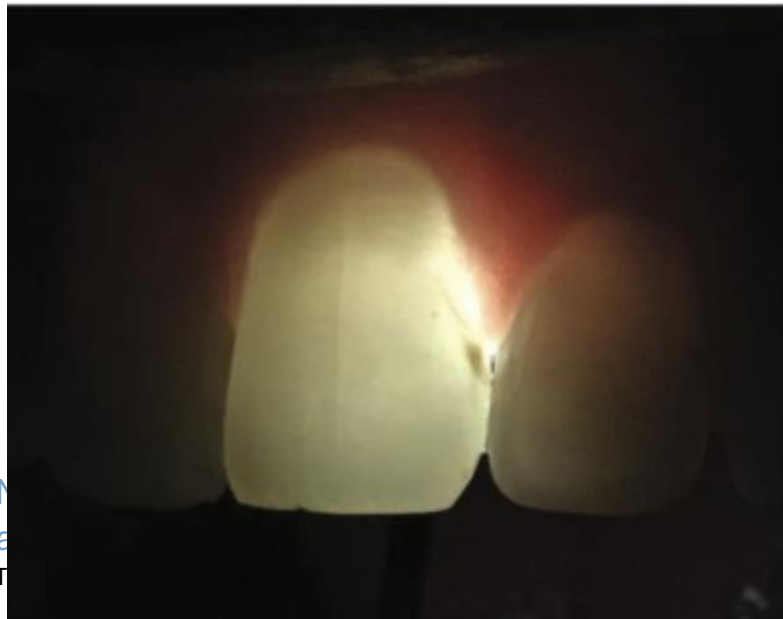


Fiber Optic Trans Illumination (FOTI)

- A narrow beam of bright white light that is directed across areas of contact between the proximal surfaces and the disruption of crystal structure that deflects the light beam and thus produces shadows



Transillumination is particularly helpful for anterior teeth, which are narrower B-L and do not allow for BW radiographs.



Transillumination can also be used to highlight cracks and their depth.

Cracks that reach the DEJ will prevent the transmission of light beyond the crack.

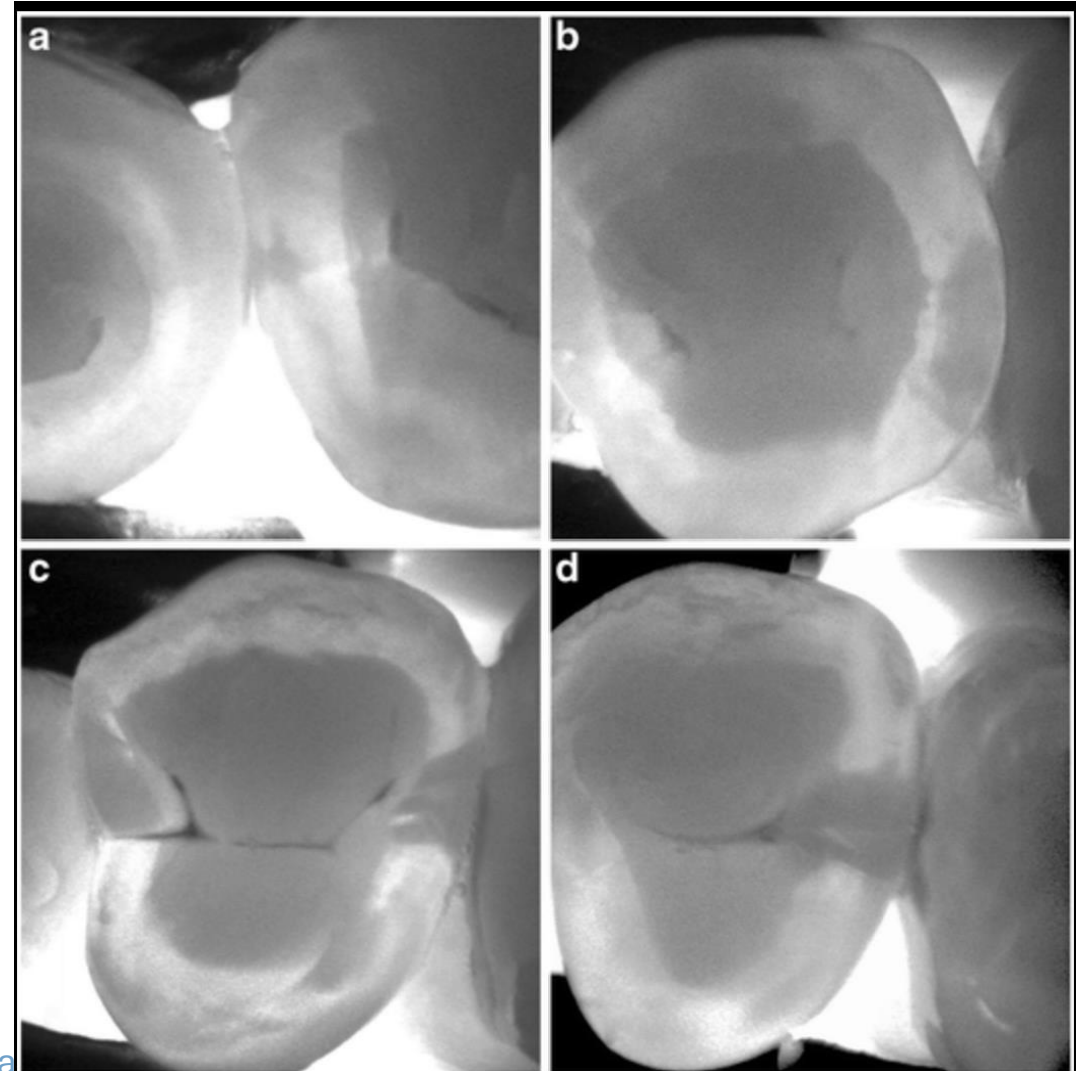


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Near Infrared Imaging

- Uses infrared (**780-1,500 nm wavelength**) transillumination as an added feature to some intraoral scanners
- Infrared has better contrast and is easier to photograph than visible light.
- Caries appears dark due to difference in refraction of demin structure

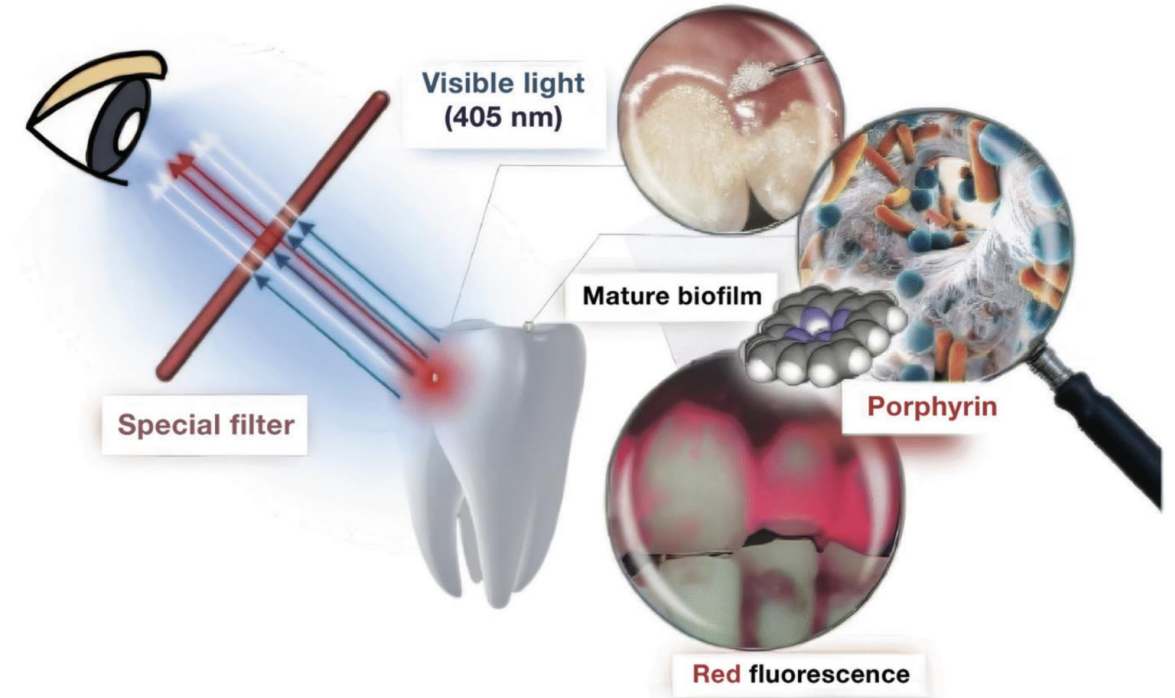


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Fluorescence in Caries Detection

- Fluorescence: Absorbing a short wavelength of light and emitting a longer wavelength
- Dental fluorescence devices typically use a **405nm blue-violet light**
- **Healthy enamel fluoresces brightly in green/yellow colors**
- **Demineralized enamel = less fluorescent/darker**
- **Oral microbial metabolites = emit red fluorescence (RF)**
 - o From porphyrin induced metabolites produced by oral microbes
 - o Can be caries or can be plaque/calculus



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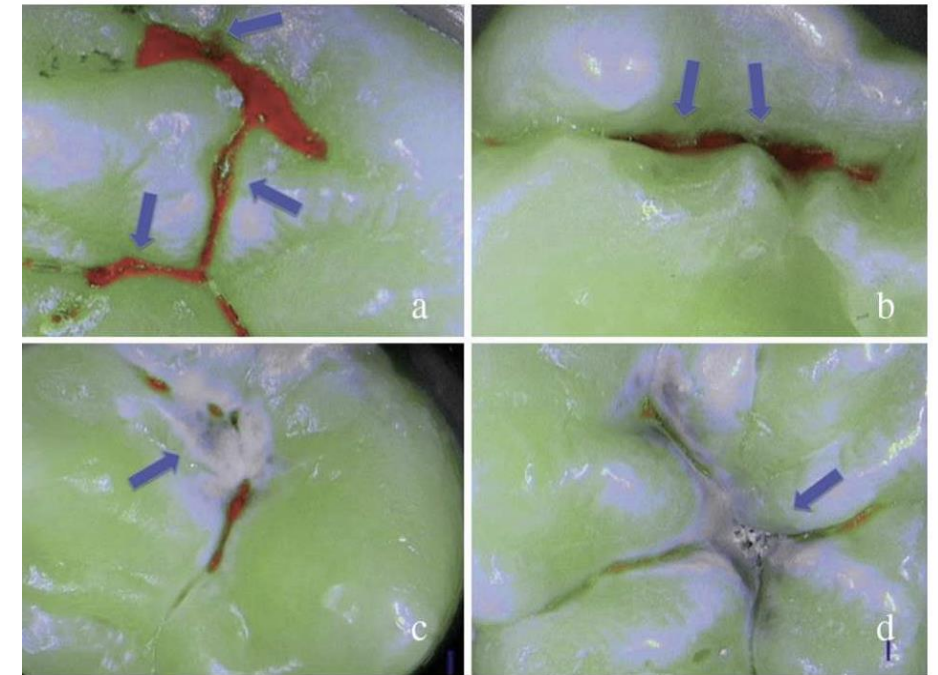
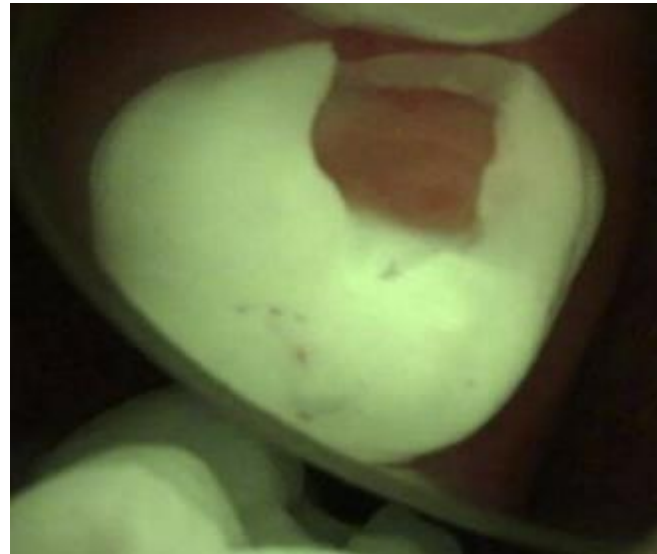
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Fluorescence Cameras

Most common devices are dual purpose intraoral cameras that take both normal white light photographs and fluorescent photographs.

They may also suggest a scale of codes that correlate with the manufacturers use and treatment recommendations.

- **Red fluorescence** → **presence of bacterial metabolites**
- **Fluorescence loss** → **demineralization**



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Quantitative Light Fluorescence

- More common in research. Most clinical fluorescent cameras don't give true quantitative results.
- **Quantitative devices measures loss of fluorescence in the enamel/dentin**
 - **Can produce a number (ΔF), showing how much fluorescence was lost**
 - % loss of fluorescence is about equal to the % loss of mineral and depth of lesion.
 - **Can also produce a number (ΔR), showing microbial (lesion) activity.**
 - Clinical experiencesuggests that RF correlates with the presence of mature plaque and calculus. Therefore, a demineralized site with RF can be presumed an active lesion.

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Table 1. Quantitative parameters obtained with QLF

Name	Symbol	Unit	Description
Delta F	ΔF	%	Percentage fluorescence loss with respect to the fluorescence of sound tooth tissue. Related to lesion depth.
Delta Q	ΔQ	%px ²	Percentage fluorescence loss with respect to the fluorescence of sound tissue times the area. Related to lesion volume.
Lesion Area	$A_{\Delta F}$	px ²	Area with ΔF equal or smaller than a specific threshold value of ΔF (default -5%).
Delta R	ΔR	%	Percentage of increase of the ratio of the red and the green component with respect to that ratio of sound tissue. Related to the presence of porphyrins and indirectly related to bacterial activity.
RF Area	$A_{\Delta R}$	px ²	Area with ΔR equal or higher than a specific threshold value of ΔR.
Simple Plaque Score™	SPS™	-	A value from 0 (no mature plaque) to 5 (high amount of mature plaque).
Two-Tone Plaque Score™	TTPS™	%	Percentages of tooth area covered by mature (dark-blue) and immature plaque (pink-blue).

Table 2. ICQ 2011 in vivo Index Manal Allammari et al. 2010

QLF SCORE	ICDAS SCORE	DESCRIPTION	ΔF	ΔR
0	0	Sound tooth surface	-0,5 .. -12	0 .. 20
1	1	First visual change in enamel	-12,5 .. -18	21 .. 35
2	2	Distinct visual change in enamel	-18,5 .. -26	36 .. 60
3	3	Localized enamel breakdown because of caries with no visible dentin or underlying shadow	-26,5 .. -32	61 .. 88
	4	Underlying dark shadow from dentin with or without localized enamel breakdown	-32,5.. -38	89 .. 95
4	5	Distinct cavity with visible dentin	-38,5 .. -48	96 .. 110
	6	Extensive distinct cavity with visible dentin	> 48,5	> 110

- Red fluorescence (RF) can be used as an indicator of microbiologic activity
- Caries detector during excavation
 - Plaque disclosing agent during hygiene visits
 - Helping to differentiate stain vs caries (if tooth is thoroughly cleaned first)

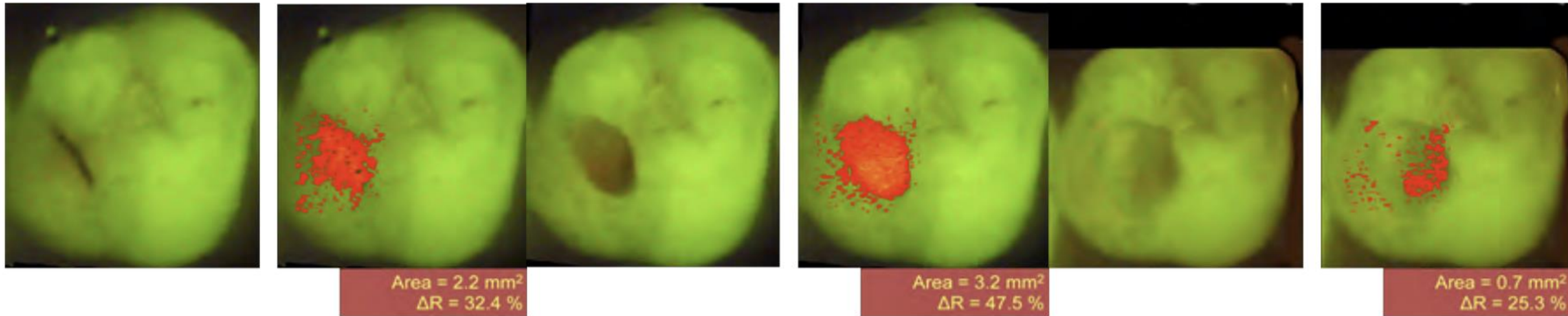


Figure 14. Use of QLF™ during excavation (images made with the Inspektor™ Pro). From left to right: Discolored fissure that shows RF; After opening the fissure a dentine lesion is exposed with corresponding increase in RF. The excavation prior to preparation. Some RF is still present (*Images made by P.Sas, Amsterdam, Netherlands*).

remember what these devices are actually detecting.
RF margins can be either recurrent caries or can be the accumulation of plaque at restorative margins (without recurrent caries).



Figure 2. Example of the images taken semi-simultaneously with the QLF-D Biluminator™ 2. To the left the 'normal' white light image. In the middle the corresponding QLF™-image. Note the absence of reflections and the red fluorescence in the QLF™-image. Also note the correspondence of the areas with RF and plaque seen on the white light image.



Figure 16. Example of secondary caries. The QLF™-image shows a lot of RF along the margins of the fillings. The left of the two central elements shows demineralization in combination with RF extending from the filling margin at the lower end of the surface (*images courtesy of Karen van Daelen, SWT-TN*).

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Diagnodent

- **Measures fluorescence produced by bacterial metabolites like porphyrins**
- **Emits a 655nm wavelength diode laser**
 - Light is absorbed, reflected back, and quantified by the machine on a scale of 0-99
 - Higher number = more porphyrins (more red)
 - However NO relationship between the number and the DEPTH of decay
 - Most practitioners agree 30-35 is reflective of dentinal involvement.
 - But what is a diagnodent actually measuring? Is it the depth of the lesion? The degree of demineralization?
- **Baseline fluorescence can vary between patients**
 - Take a baseline reading on clean, dry, healthy smooth surface enamel for each patient first.
- **Most useful for trends and activity in pit and fissure lesions**
 - If trending downwards over time, may be inactive
 - If trending upwards over time, may be active



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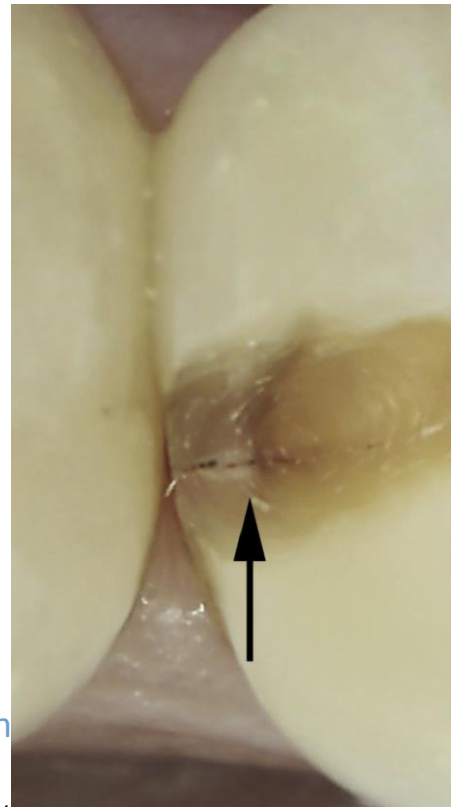
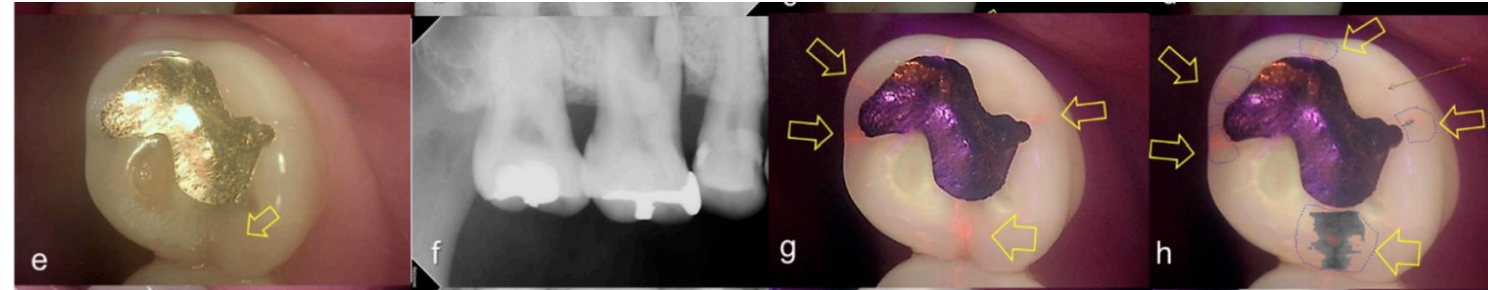
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Caries in Cracks

Cracks can allow bacterial ingress and become protected sites. However, not all discolored cracks are treated the same.

Assess for:

- Signs of bioactivity
 - Stick to explorer
 - RF/Diagnodent
- Depth and width
 - Dentin visibly/radiographically involved
 - Pain on biting/release
 - Fits explorer or widening over time
- Structural prognosis based on patient risk factors and history
 - Partial crack removal, like partial caries removal, can sometimes suffice to control future disease if etiologic factors are well managed



- Assistive diagnostic technologies tend towards false positives if used as a primary tool. However, as an adjunct, they can help you assess activity and trends. They may also aid in new patient communication and education.
- Bitewing radiographs are the most diagnostic radiographs for caries.
- If a D3 lesion is appreciable on a BW, a PA is recommended too.
- Recommended radiographic frequency is based on the patient's individual risk category.
- Understanding etiology aids in both accurate diagnosis and preventive prognosis.
- **Visual-tactile exam of a clean, dry, well lit tooth, with consideration for your patient's history and risk factors, is the goldstandard and most accurate among experienced dentists.**
- Only a dentist can diagnose a dental condition.